

Multi-Antenna UAV Extension for Active STAR-RIS Assisted UAV Networks With NOMA

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This supplementary note presents a multi-antenna UAV extension of the proposed active STAR-RIS assisted UAV network with NOMA. In the main paper, the UAV is equipped with a single antenna in order to preserve the tractability of the analytical derivation. To further illustrate the scalability of the proposed framework, we consider a multi-antenna UAV case, where the UAV is equipped with $\mathcal{M}$ antennas. It should be noted that the antenna selection problem and beamforming optimization are not further considered in this supplementary note.

I. SIGNAL MODEL FOR MULTI-ANTENNA UAV CASE

Let $\mathcal{M}$ be the number of antennas of UAV, then the received signal can be denoted as

$$Y_\kappa = \rho \sum_{m=1}^{\mathcal{M}} (\mathbf{h}_m \Theta_\kappa \mathbf{g}_{m,\kappa} \Phi_\kappa X + \Theta_\kappa \mathbf{g}_{m,\kappa} \mathbf{n}_1) + n_0 \quad (1)$$

Further extending the above formula, we can derive the SINR expressions of active STAR-RIS assisted multi-antenna UAV networks with NOMA, i.e.,

$$\gamma_s = \frac{\rho^2(1-a)P\Phi_2^2|\chi_{2,\mathcal{M}}|^2}{\rho^2(aP\Phi_2^2|\chi_{2,\mathcal{M}}|^2 + \sigma_1^2 d_{s2}^{-\alpha_2} |G_{2,\mathcal{M}}|^2) + P\sigma_e^2 + \sigma_0^2} \quad (2)$$

$$\gamma_2 = \frac{\rho^2 a P \Phi_2^2 |\chi_{2,\mathcal{M}}|^2}{\rho^2 \sigma_1^2 d_{s2}^{-\alpha_2} |G_{2,\mathcal{M}}|^2 + \beta_{RI} \mathbb{A} + (1 + \beta_{RI}) a P \sigma_e^2 + \sigma_0^2} \quad (3)$$

$$\gamma_1 = \frac{\rho^2(1-a)P\Phi_1^2|\chi_{1,\mathcal{M}}|^2}{\rho^2(aP\Phi_1^2|\chi_{1,\mathcal{M}}|^2 + \sigma_1^2 d_{s1}^{-\alpha_2} |G_{1,\mathcal{M}}|^2) + P\sigma_e^2 + \sigma_0^2} \quad (4)$$

where $\chi_{\kappa,\mathcal{M}} = \sum_{m=1}^{\mathcal{M}} \mathbf{h}_m \Theta_\kappa \mathbf{g}_{m,\kappa}$ and $G_{\kappa,\mathcal{M}} = \sum_{m=1}^{\mathcal{M}} \Theta_\kappa \mathbf{g}_{m,\kappa}$.

To calculate the analytical expressions of downlink metrics for active STAR-RIS assisted multi-antenna UAV networks with NOMA, we need to re-define $|\chi_{\kappa,\mathcal{M}}|^2$, then using the derivation method provided by our work.

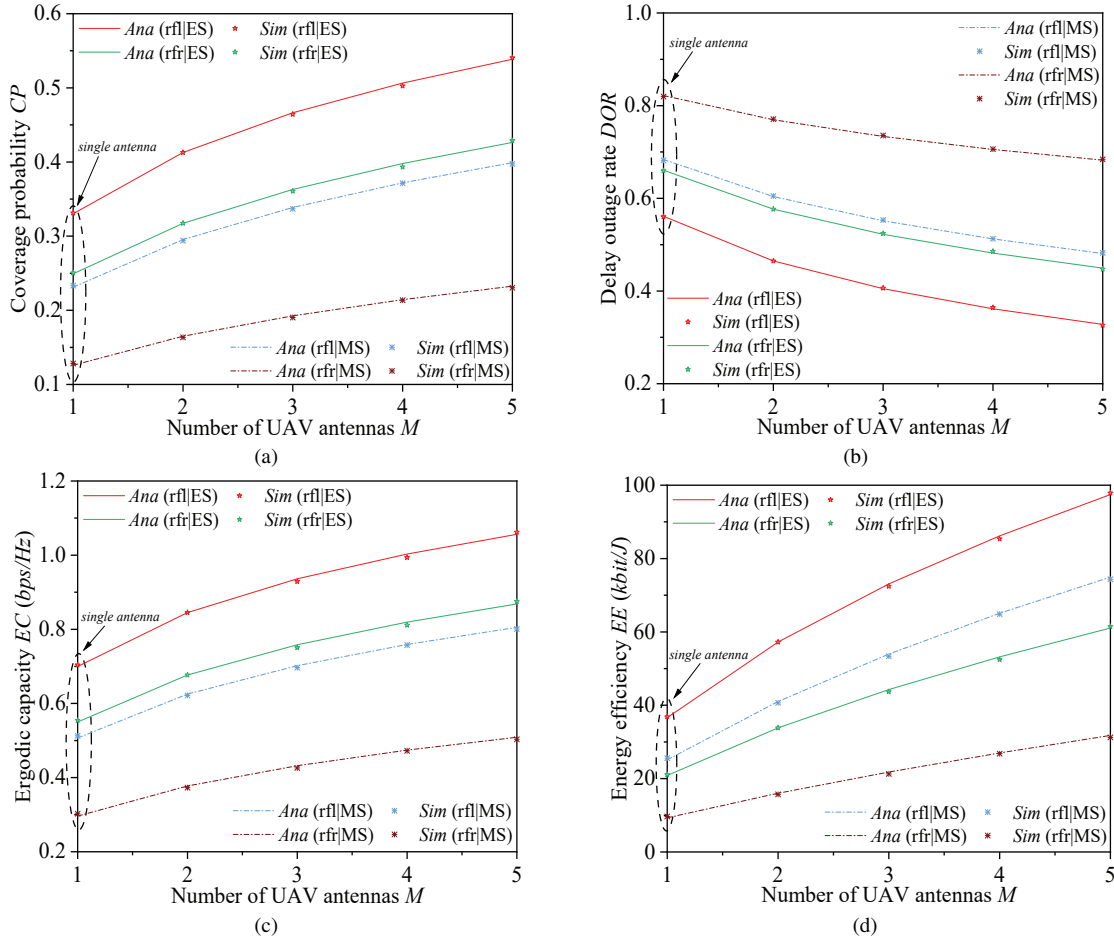
II. PERFORMANCE EVALUATION

Using the same parameters setting of the second experiment group ($N = 200$, Section III.B), fig. 1 exhibits the influence of number of antennas of UAV on the downlink performance of active STAR-RIS assisted multi-antenna UAV networks with

NOMA (for $\mathcal{M} = 1$, it becomes the proposed model of our article).

It is easy to find that with the increase of number of antennas, the downlink performance of active STAR-RIS assisted multi-antenna UAV network with NOMA would be improved. The reason is simple, i.e., the increasing number of antennas would of course enhance the AGI signal quality. Similarly, $\mathcal{U}_2$ attains better downlink performance than $\mathcal{U}_1$ and executing ES protocol yields superior performance compared to executing MS protocol (analyzed in our article). However, it should be also noted that adding the number of antennas would result in the increase of hardware costs. Due to the finite size of UAV, it is also impossible for us to infinitely add the number of antennas in practice (that's the reason that new antenna techniques are proposed in recent years, such as fluid antenna, continuous antenna, pinching antenna and etc).

It should also be noted that adding the number of UAV antennas would increase the hardware cost, circuit power consumption and payload burden of UAV. Due to the finite size and limited energy supply of UAV, it is not practical to infinitely increase the number of antennas in real deployment. Therefore, the multi-antenna extension mainly demonstrates the potential performance improvement brought by antenna diversity, while practical antenna configuration and antenna selection optimization are left for future work.

Fig. 1. The impact of number of UAV antennas $\mathcal{M}$